

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO5: Apply computer vision techniques to real-world applications such as face recognition, tracking, medical imaging, and autonomous systems. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 20 Total Marks:20 Annexure A

Mock Interview

Code of Conduct:

I JAASIEL J S (192524046) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge	
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem solving ability; requires guidance	Unable to solve or apply concepts	
Analytical Thinking	15	Analyzes questions critically and provides well structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated	
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively	

Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism	
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately	

Annexure A

Mock Interview

1. What is Eigenfaces?

ANS: Eigenfaces is a face recognition method based on Principal Component Analysis (PCA) that represents faces using a set of important eigenvectors called Eigenfaces.

2. Define PCA in image processing.

ANS: Principal Component Analysis (PCA) is a dimensionality reduction technique that transforms image data into a smaller set of important features while preserving maximum variance.

3. What is contour-based shape representation?

ANS: Contour-based shape representation describes an object using its boundary or outline, focusing on the external shape of the object.

4. What is region-based shape representation?

ANS: Region-based shape representation describes an object using all pixels inside its region, including its area, texture, and internal properties.

5. How does an active contour (snake) work?

ANS: An active contour starts with an initial curve and iteratively moves it toward the object boundary by minimizing an energy function based on internal smoothness and image features.

6. How does a level set method work?

ANS: A level set method represents a contour as a level curve of a higher-dimensional function and evolves the contour to detect and segment object boundaries.

7. What is the Hough Transform?

ANS: The Hough Transform is a feature extraction technique used to detect geometric shapes such as lines and circles by transforming image points into a parameter space.

8. How are Fourier descriptors used in shape analysis?

ANS: Fourier descriptors represent an object's boundary using Fourier coefficients, which can be used to describe and compare shapes.

9. How are wavelet descriptors used in shape analysis?

ANS: Wavelet descriptors represent shapes at different scales and frequencies, allowing both global and local shape features to be analyzed.

10. What is a particle filter?

ANS: A particle filter is a probabilistic tracking method that uses multiple particles to represent possible states of an object and updates them using motion and observation information.

11. How is foreground–background separation done in surveillance systems?

ANS: Foreground–background separation is performed by modeling the background and detecting changes or moving regions as foreground objects using techniques such as background subtraction.

12. How do multi-camera systems track pedestrians?

ANS: Multi-camera systems detect pedestrians in each camera and use tracking and person re-identification features to maintain the same identity across different camera views.

13. What is a Convolutional Neural Network (CNN)?

ANS: A CNN is a deep learning network that uses convolutional layers to automatically extract spatial features from images for tasks such as classification and detection.

14. How does a Vision Transformer (ViT) work?

ANS: A Vision Transformer divides an image into patches, converts them into embeddings, and uses Transformer self-attention to learn relationships between the image patches.

15. What is a Generative Adversarial Network (GAN)?

ANS: A GAN is a generative model consisting of a Generator and a Discriminator that compete with each other to generate realistic synthetic data.

16. How is a Support Vector Machine (SVM) used for image classification?

ANS: An SVM classifies images by finding an optimal decision boundary that separates image features into different classes with maximum margin.

17. How are Hidden Markov Models (HMM) applied in activity recognition?

ANS: HMMs model activities as sequences of hidden states and use observed features over time to recognize and classify human activities.

18. What is a Gaussian Mixture Model (GMM)?

ANS: A GMM is a probabilistic model that represents data as a mixture of multiple Gaussian distributions and is commonly used for clustering and background modeling.

19. How is graph-based learning applied to 3D shape correspondence?

ANS: Graph-based learning represents 3D shapes as graphs and learns relationships between their vertices or features to establish

correspondence between similar shapes.

20 How is road and lane detection performed in autonomous driving?

ANS: Road and lane detection is performed using image preprocessing, edge detection, region-of-interest selection, and techniques such as Hough Transform or deep learning to identify road and lane boundaries.